

TaloNet - Bill of Materials & Procurement

Peregrine net-interceptor drone + ForensIQ-1 forensic appliance.

One section per subsystem; Ext = Qty x Unit; each section subtotals; the Cost Roll-up sums everything with a 15% contingency. Prices are INDICATIVE USD - confirm with the vendor before ordering. Build/assembly: docs/12_Garage_Build_Guide.md. Source: tools/generate_bom.py.

Disposition legend

MAKE	fabricate/assemble in the garage
BUY	purchase off-the-shelf
OUTSOURCE	send a drawing/gerber out (CNC, PCB, finish)
TOOL	one-time shop equipment

SAFETY/LEGAL: high-energy LiPo, pressurized CO2 and a kinetic net launcher are no joke - see docs/12 safety. Authorized operators only; intelligence + human-in-the-loop trigger, never an autonomous weapon.

Drone - Airframe & Propulsion

ID	Item / Description	Mfr / Part No.	Qty	Disp	Unit\$	Ext\$	Vendor	Notes
DR-AF-001	Hex carbon deck plate, 360 mm AF, 3 mm (top/bottom)	TaloNet PG-DECK (CNC)	2	OUTSOURCE	\$130	\$260	Local CNC shop	From talonet_frame.scad; CF/G10,...
DR-AF-002	Carbon arm tube 30x26 mm, 420 mm	Rock West 30x26	4	BUY	\$22	\$88	Rock West Composites	Roll-wrapped CF; X8 = 4 arms
DR-AF-003	Arm clamp + coaxial motor mount (alu 6061)	TaloNet PG-MNT (CNC)	4	OUTSOURCE	\$38	\$152	Local CNC shop	Holds 2 coaxial motors/arm
DR-AF-004	Standoff/spacer set, alu, deck stack	M3 alu standoff kit	1	BUY	\$18	\$18	McMaster-Carr	deck_gap 70 mm
DR-AF-005	Landing gear carbon arch tube 22 mm + alu feet	TaloNet PG-GEAR	2	OUTSOURCE	\$45	\$90	Local CNC shop	gear_height 230 mm
DR-PR-001	Brushless motor, 14 kg-class octo, ~KV120	T-Motor MN801S KV120	8	BUY	\$175	\$1,400	T-Motor	X8 coaxial; 8 motors total
DR-PR-002	Folding CF prop ~28-30 in (CW/CCW pair)	T-Motor NS28x9.2 CF	8	BUY	\$46	\$368	T-Motor	prop_dia ~711 mm
DR-PR-003	ESC 80 A 12S HV, DShot600 + telemetry	T-Motor FLAME 80A 12S	8	BUY	\$95	\$760	T-Motor	1 per motor
SUBTOTAL						\$3,136		

Drone - Avionics, Power & C2

ID	Item / Description	Mfr / Part No.	Qty	Disp	Unit\$	Ext\$	Vendor	Notes
DR-AV-001	Flight controller, triple-IMU + carrier	CubePilot Cube Orange+	1	BUY	\$430	\$430	CubePilot / IRLock	ArduPilot; SERVO/RELAY outputs
DR-AV-002	GNSS receiver w/ Galileo OSNMA	Septentrio mosaic-X5 board	1	BUY	\$620	\$620	Septentrio	anti-spoof authenticated nav
DR-AV-003	Secondary GNSS/compass (CAN)	Hex Here3+	1	BUY	\$120	\$120	CubePilot	RAIM cross-check
DR-AV-004	Companion compute module + carrier	NVIDIA Jetson AGX Orin 32GB	1	BUY	\$1,199	\$1,199	NVIDIA / Seeed	EO/IR pipeline + defense/ ; VLM re...
DR-AV-005	Long-range telemetry radio pair, 900 MHz	RFD900x (pair)	1	BUY	\$200	\$200	RFDDesign	MAVLink2 link to GCS
DR-AV-006	EO/IR gimbal camera (FPV feed)	Workswell/SIYI ZT6 EO+IR	1	BUY	\$950	\$950	SIYI	RTSP/UDP to gcs cockpit
DR-PW-001	12S6P Li-ion / LiPo pack ~22 Ah	Tattu 12S 22000 mAh	1	BUY	\$520	\$520	Tattu / GensAce	endurance/range driver
DR-PW-002	Power module / current sensor 200 A	Mauch HS-200-HV + PowerCube	1	BUY	\$95	\$95	Mauch	I2C power monitor
DR-PW-003	5.3 V/8 A BEC (redundant, OR-ing)	Mauch 5.3V/8A BEC	2	BUY	\$28	\$56	Mauch	avionics bus
DR-PW-004	Main connectors + precharge (AS150/XT90-S)	AS150 + XT90-S set	1	BUY	\$22	\$22	Amass	inrush limit
DR-PW-005	Fuse set: ANL 150 A + blade 3-10 A	Littelfuse ANL/ATO kit	1	BUY	\$24	\$24	Littelfuse	per wire/fuse sizing
SUBTOTAL						\$4,236		

Drone - Net Payload

ID	Item / Description	Mfr / Part No.	Qty	Disp	Unit\$	Ext\$	Vendor	Notes
DR-NP-001	Digital servo 20 kg (AIM-PAN, SERVO9)	Savox SB-2290SG	1	BUY	\$80	\$80	Savox	gimbal pan -60..+60 deg
DR-NP-002	Digital servo 20 kg (AIM-TILT, SERVO10)	Savox SB-2290SG	1	BUY	\$80	\$80	Savox	gimbal tilt 0..75 deg
DR-NP-003	Pan/tilt gimbal yoke + bearing (printed/CNC)	TaloNet PG-GIMBAL	1	MAKE	\$35	\$35	Garage 3D print	net_launcher(); PETG-CF/alu
DR-NP-004	Servo UBEC 6 V / 5 A	Castle 10A BEC Pro	1	BUY	\$22	\$22	Castle Creations	servo bus, isolated
DR-NP-005	Brushed gearmotor 12 V (mouth cinch)	Pololu 25D HP 12V	1	BUY	\$40	\$40	Pololu	drawstring spool; SERVO11 driver
DR-NP-006	Motor driver H-bridge w/ current sense	Pololu G2 24v13	1	BUY	\$30	\$30	Pololu	cinch drive
DR-NP-007	Current sensor breakout	INA226 module	1	BUY	\$8	\$8	Pololu/Adafruit	tension/stall threshold
DR-NP-008	Drawstring spool + flanges (printed/turned)	TaloNet PG-SPOOL	1	MAKE	\$12	\$12	Garage 3D print	cinch_mechanism()
DR-NP-009	Brushed gearmotor 12 V (hoist winch)	Pololu 37D 12V	1	BUY	\$40	\$40	Pololu	vertical recovery
DR-NP-010	Motor driver H-bridge 21 A	Pololu G2 24v21	1	BUY	\$40	\$40	Pololu	winch drive
DR-NP-011	Load cell 5 kg + HX711 ADC	TAL220 + HX711	1	BUY	\$11	\$11	SparkFun	winch tension feedback
DR-NP-012	CO2 launcher: solenoid valve + 16 g cartridge	12V NC CO2 solenoid	1	BUY	\$35	\$35	Industrial supply	RELAY0 fired
DR-NP-013	Logic-level MOSFET (low-side switch)	Infineon IRLB3034PBF	2	BUY	\$2	\$4	Mouser	solenoid/relay drive
DR-NP-014	Arming relay + flyback diode set	Omron G5LE + 1N5408	1	BUY	\$9	\$9	Mouser	arming interlock
DR-NP-015	Quick-release servo (drop/abort)	Savox SB-2290SG	1	BUY	\$80	\$80	Savox	SERVO12 release
DR-NP-016	Capture net, knotless Dyneema/Kevlar, ~3.5 m mouth	Custom cast net	1	MAKE	\$60	\$60	Garage / net maker	capture_net(); launched casting net
DR-NP-017	Drawstring (Dyneema 2 mm) + perimeter weights	Dyneema + brass weights	1	MAKE	\$18	\$18	Garage	purse loop + rim weights
DR-NP-018	Muzzle nozzle + bay housing (printed)	TaloNet PG-MUZZLE	1	MAKE	\$20	\$20	Garage 3D print	netlauncher_bay()
DR-NP-019	Payload interface PCB (STM32G474) fab + assembly	TaloNet PG-PAYIF rev A	1	OUTSOURCE	\$65	\$65	JLCPCB/PCBWay	interlock + driver carrier
SUBTOTAL						\$689		

Forensic Appliance (ForensIQ-1)

ID	Item / Description	Mfr / Part No.	Qty	Disp	Unit\$	Ext\$	Vendor	Notes
FA-CP-001	Raspberry Pi Compute Module 4 (4GB/32GB)	Raspberry Pi CM4104032	1	BUY	\$75	\$75	PiHut / DigiKey	runs forensics/
FA-CP-002	CM4 carrier board (USB3, NVMe, GbE, DSI)	Waveshare CM4-IO-BASE-B	1	BUY	\$45	\$45	Waveshare	I/O for appliance
FA-UI-001	7" capacitive touchscreen, 1024x600 DSI	Waveshare 7inch DSI LCD	1	BUY	\$70	\$70	Waveshare	operator UI
FA-WB-001	Hardware USB write-blocker (forensic)	CRU WiebeTech USB WriteBloc...	1	BUY	\$350	\$350	CRU Inc.	trust anchor; read-only
FA-WB-002	USB SD/microSD reader (behind blocker)	Kingston MobileLite Plus	1	BUY	\$20	\$20	Kingston	card intake
FA-PR-001	80 mm panel-mount ESC/POS thermal printer	Rongta RP05 / CSN equiv	1	BUY	\$60	\$60	Rongta	report output
FA-PR-002	80 mm thermal paper roll (pack of 4)	Thermal 80x50 roll x4	1	BUY	\$12	\$12	Office supply	report media
FA-ST-001	NVMe SSD 512 GB (working/case store)	Samsung 980 512GB	1	BUY	\$50	\$50	Samsung	LUKS-encrypted
FA-PW-001	UPS HAT / DC-UPS + 18650 cells	Geekworm X1200 + 2x 18650	1	BUY	\$45	\$45	Geekworm	clean shutdown / field run
FA-PW-002	AC/DC inlet + 12V PSU	IEC inlet + 12V/5A PSU	1	BUY	\$22	\$22	Mean Well	mains
FA-RT-001	RTC module (UTC time source)	DS3231 + coin cell	1	BUY	\$6	\$6	Adafruit	trusted timestamps
FA-EN-001	Rugged khaki enclosure (printed/CNC) + key-lock	forensic_appliance.scad	1	MAKE	\$80	\$80	Garage 3D print	ForensIQ-1 case; tamper lock
FA-EN-002	Status LEDs, START/EJECT buttons, key-lock	Panel control kit	1	BUY	\$30	\$30	Adafruit/AliExpress	front panel
FA-EX-001	Write-once / WORM USB (evidence export)	Kanguru WORM USB 32GB	1	BUY	\$40	\$40	Kanguru	evidentiary PDF/map
FA-CL-001	60 mm fan + filtered intake	Noctua NF-A6x25 5V	1	BUY	\$16	\$16	Noctua	thermal
SUBTOTAL						\$921		

Fasteners, Wiring & Consumables

ID	Item / Description	Mfr / Part No.	Qty	Disp	Unit\$	Ext\$	Vendor	Notes
CO-WR-001	Silicone wire kit 10/12/14/18/20/22 AWG	BNTECHGO silicone kit	1	BUY	\$40	\$40	BNTECHGO	per fuse/AWG sizing
CO-WR-002	XT60/XT90/JST-GH/Molex + crimp pins	Amass + JST kit	1	BUY	\$35	\$35	Amass / JST	looms
CO-WR-003	Heat shrink + braided sleeve + Kapton	Assorted shrink kit	1	BUY	\$18	\$18	3M	harness
CO-FS-001	M2/M2.5/M3 stainless socket-head + nylocs	McMaster screw assortment	1	BUY	\$45	\$45	McMaster-Carr	airframe + payload
CO-AD-001	Threadlocker, CA, 30-min epoxy, RTV	Loctite 243 + JB Weld	1	BUY	\$30	\$30	Loctite	vibration-proofing
CO-3D-001	PETG-CF / ASA filament (printed parts)	Prusament PETG-CF 1kg	2	BUY	\$40	\$80	Prusa	gimbal/spool/muzzle/enclosure
CO-TH-001	Thermal pads/paste + standoffs (Jetson/CM4)	Thermal kit	1	BUY	\$15	\$15	Arctic	compute cooling
SUBTOTAL						\$263		

ID	Item / Description	Mfr / Part No.	Qty	Disp	Unit\$	Ext\$	Vendor	Notes
OS-CN-001	CNC/waterjet carbon decks (2x) + alu mounts	per cad STEP/STL	1	OUTSOURCE	\$0	\$0	Local CNC / SendCutSend	covered in DR-AF lines; quote per ...
OS-PC-001	Payload IF PCB fab + SMT assembly (STM32G474)	TaloNet PG-PAYIF rev A	1	OUTSOURCE	\$0	\$0	JLCPCB / PCBWay	covered in DR-NP-019; provide ge...
OS-AN-001	Anodize/powder-coat alu parts (khaki #86855F)	finish service	1	OUTSOURCE	\$60	\$60	Local finisher	signature khaki
SUBTOTAL						\$60		

Shop Tools & Equipment

ID	Item / Description	Mfr / Part No.	Qty	Disp	Unit\$	Ext\$	Vendor	Notes
TL-001	Soldering station + hot air rework	Hakko FX-951 / 8586	1	TOOL	\$130	\$130	Hakko	harness + payload IF
TL-002	Ratchet crimpers (XT/JST/Molex/ferrule)	IWISS crimp set	1	TOOL	\$70	\$70	IWISS	connectors
TL-003	FDM 3D printer (enclosed, CF-capable)	Bambu Lab X1C / Prusa MK4	1	TOOL	\$1,200	\$1,200	Bambu Lab	gimbal/spool/muzzle/enclosure (C...
TL-004	Digital calipers + bench multimeter	Mitutoyo + Fluke 117	1	TOOL	\$180	\$180	Mitutoyo/Fluke	QA
TL-005	Bench PSU 30V/10A + LiPo charger	Riden + ISDT	1	TOOL	\$150	\$150	Riden / ISDT	bring-up + battery
TL-006	Torque drivers, hex set, helping hands, heat gun	Wera + misc	1	TOOL	\$120	\$120	Wera	assembly
TL-007	LiPo-safe bag/bunker, CO2-safe handling, fume ext.	Safety kit	1	TOOL	\$90	\$90	misc	battery + CO2 + solder fumes
SUBTOTAL						\$1,940		

Cost Roll-up

Drone - Airframe & Propulsion	\$3,136
Drone - Avionics, Power & C2	\$4,236
Drone - Net Payload	\$689
Forensic Appliance (ForensIQ-1)	\$921
Fasteners, Wiring & Consumables	\$263
Outsourced Fabrication	\$60
Shop Tools & Equipment	\$1,940
Materials subtotal (incl. tools)	\$11,245
Contingency 15%	\$1,687
GRAND TOTAL (indicative)	\$12,932